

Whiteness-based bilevel estimation of weighted TV parameter maps for image denoising

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Abstract. We consider a bilevel optimisation strategy based on normalised residual whiteness loss for estimating the weighted total variation parameter maps for denoising images corrupted by additive white Gaussian noise. Compared to supervised and semi-supervised approaches relying on prior knowledge of (approximate) reference data and/or information on the noise magnitude, the proposal is fully unsupervised. To avoid noise overfitting an early stopping strategy is used, relying on simple statistics of optimal performances on a set of natural images. Numerical results comparing the supervised/unsupervised procedures for scalar/pixel-dependent parameter maps are shown.

Keywords: Adaptive regularisation · Total variation image denoising · Unsupervised bilevel optimisation · Residual whiteness.

1 Introduction

Variational methods are a popular paradigm to overcome the difficulties related to inverse imaging problems: they seek an estimate of the unknown image by minimizing a functional encoding information on the noise and prior beliefs on the image. For image denoising, denoting by $\mathbf{y} \in \mathbb{R}^n$ the (vectorised) data corrupted by Additive White Gaussian Noise (AWGN), variational methods solve

$$\mathbf{x}^*(\boldsymbol{\lambda}) \in \arg \min_{\mathbf{x} \in \mathbb{R}^n} \left\{ \frac{1}{2} \|\mathbf{x} - \mathbf{y}\|_2^2 + R(\mathbf{x}; \boldsymbol{\lambda}) \right\},$$

where $\mathbf{x}^*$ depends on a certain number of parameters $\boldsymbol{\lambda} \in \Lambda$ possibly involved in the so-called regularisation term R . In general, flexible models are characterised by a large number of parameters for which robust selection strategies need to be introduced. How to properly select parameters in variational models is an issue that has attracted the interest of many researchers working on inverse problems.